

11 Group 17

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Halogen oxidising strength decreases down the group; halide reducing ability increases. A more reactive halogen displaces a less reactive halide from solution. Halide ions give silver halide precipitates with AgNO_3 after acidification; colour and ammonia solubility help identify ions.

Worked example: follow the reasoning

Cl_2 oxidises Br^- : $\text{Cl}_2 + 2\text{Br}^- \rightarrow 2\text{Cl}^- + \text{Br}_2$; chlorine is reduced and bromide oxidised.

Quick check · multiple choice

A Which halogen oxidises Br^- to Br_2 in aqueous solution?

[1]

- A I_2
- B Br_2
- C Cl_2
- D Br^-

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Predict observation when chlorine water is added to aqueous potassium iodide, and write ionic equation.

[3]

Answers and explanation

Quick check: C. Cl₂

Question 1 [3 marks]

Brown iodine appears [1]; $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$ [2].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.